Time limits. Each instance has a time limit of 300 seconds: 180 seconds for the exact search and up to 120 seconds for a fallback heuristic. The heuristic may terminate early, so runs without proven optimality can finish in slightly more than 180 seconds.

1. Detailed Results of Uniform-Weight Instances on S_A under WTFRPA Setting

We first report the results of Ha26 on the instance with $ws = 0.3$. Among these instances, only the instance "60-10-sif-0.3-3" is feasible.

Table 1 Detailed Results of Ha26 on 60-10-sif-0.3-3

ws-0.3			
Name	Cost	Gap‰	CPU
60-10-sif-3	1081.0	0.00	13.08

1.1. Results on the instances with i execution mode

Table 2 Detailed Results of Ha26 on Uniform-Weight Instances in S_A with i execution mode ($ws = 0.4 - 0.6$)

ws-0.4				ws-0.5				ws-0.6			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-i-1	-	-	-	60-10-i-1	-	-	-	60-10-i-1	-	-	-
60-10-i-2	-	-	-	60-10-i-2	-	-	-	60-10-i-2	-	-	-
60-10-i-3	-	-	-	60-10-i-3	-	-	-	60-10-i-3	1082.0	0.00	11.00
60-10-i-4	-	-	-	60-10-i-4	-	-	-	60-10-i-4	-	-	-
60-10-i-5	-	-	-	60-10-i-5	-	-	-	60-10-i-5	-	-	-
60-20-i-1	-	-	-	60-20-i-1	-	-	-	60-20-i-1	-	-	-
60-20-i-2	-	-	-	60-20-i-2	-	-	-	60-20-i-2	-	-	-
60-20-i-3	-	-	-	60-20-i-3	-	-	-	60-20-i-3	-	-	-
60-20-i-4	-	-	-	60-20-i-4	-	-	-	60-20-i-4	-	-	-
60-20-i-5	-	-	-	60-20-i-5	-	-	-	60-20-i-5	-	-	-
60-30-i-1	-	-	-	60-30-i-1	-	-	-	60-30-i-1	-	-	-
60-30-i-2	-	-	-	60-30-i-2	-	-	-	60-30-i-2	-	-	-
60-30-i-3	-	-	-	60-30-i-3	-	-	-	60-30-i-3	-	-	-
60-30-i-4	-	-	-	60-30-i-4	-	-	-	60-30-i-4	-	-	-
60-30-i-5	-	-	-	60-30-i-5	-	-	-	60-30-i-5	-	-	-
90-10-i-1	-	-	-	90-10-i-1	-	-	-	90-10-i-1	-	-	-
90-10-i-2	-	-	-	90-10-i-2	-	-	-	90-10-i-2	-	-	-
90-10-i-3	-	-	-	90-10-i-3	-	-	-	90-10-i-3	-	-	-
90-10-i-4	-	-	-	90-10-i-4	-	-	-	90-10-i-4	-	-	-
90-10-i-5	-	-	-	90-10-i-5	-	-	-	90-10-i-5	-	-	-
90-20-i-1	-	-	-	90-20-i-1	-	-	-	90-20-i-1	-	-	-
90-20-i-2	-	-	-	90-20-i-2	-	-	-	90-20-i-2	-	-	-
90-20-i-3	-	-	-	90-20-i-3	-	-	-	90-20-i-3	-	-	-
90-20-i-4	-	-	-	90-20-i-4	-	-	-	90-20-i-4	-	-	-
90-20-i-5	-	-	-	90-20-i-5	-	-	-	90-20-i-5	-	-	-
90-30-i-1	-	-	-	90-30-i-1	-	-	-	90-30-i-1	-	-	-
90-30-i-2	-	-	-	90-30-i-2	-	-	-	90-30-i-2	-	-	-
90-30-i-3	-	-	-	90-30-i-3	-	-	-	90-30-i-3	-	-	-
90-30-i-4	-	-	-	90-30-i-4	-	-	-	90-30-i-4	-	-	-
90-30-i-5	-	-	-	90-30-i-5	-	-	-	90-30-i-5	-	-	-
120-10-i-1	-	-	-	120-10-i-1	-	-	-	120-10-i-1	-	-	-
120-10-i-2	-	-	-	120-10-i-2	-	-	-	120-10-i-2	-	-	-
120-10-i-3	-	-	-	120-10-i-3	-	-	-	120-10-i-3	-	-	-
120-10-i-4	-	-	-	120-10-i-4	-	-	-	120-10-i-4	-	-	-
120-10-i-5	-	-	-	120-10-i-5	-	-	-	120-10-i-5	-	-	-
120-20-i-1	-	-	-	120-20-i-1	-	-	-	120-20-i-1	-	-	-
120-20-i-2	-	-	-	120-20-i-2	-	-	-	120-20-i-2	-	-	-
120-20-i-3	-	-	-	120-20-i-3	-	-	-	120-20-i-3	-	-	-
120-20-i-4	-	-	-	120-20-i-4	-	-	-	120-20-i-4	-	-	-
120-20-i-5	-	-	-	120-20-i-5	-	-	-	120-20-i-5	-	-	-
120-30-i-1	-	-	-	120-30-i-1	-	-	-	120-30-i-1	-	-	-
120-30-i-2	-	-	-	120-30-i-2	-	-	-	120-30-i-2	6796.0	0.00	9.23
120-30-i-3	-	-	-	120-30-i-3	-	-	-	120-30-i-3	6689.0	0.00	4.03
120-30-i-4	-	-	-	120-30-i-4	-	-	-	120-30-i-4	-	-	-
120-30-i-5	-	-	-	120-30-i-5	-	-	-	120-30-i-5	-	-	-

Table 3 Detailed Results of Ha26 on Uniform-Weight Instances in S_A with i execution mode ($ws = 0.7 - 0.9$)

ws-0.7				ws-0.8				ws-0.9			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-i-1	-	-	-	60-10-i-1	1043.0	0.00	9.36	60-10-i-1	1042.0	0.00	0.79
60-10-i-2	1062.0	0.00	9.08	60-10-i-2	1059.0	0.00	9.84	60-10-i-2	1056.0	0.00	0.80
60-10-i-3	1077.0	0.00	0.85	60-10-i-3	1077.0	0.00	9.19	60-10-i-3	1077.0	0.00	10.33
60-10-i-4	-	-	-	60-10-i-4	-	-	-	60-10-i-4	-	-	-
60-10-i-5	-	-	-	60-10-i-5	-	-	-	60-10-i-5	1066.0	0.00	9.83
60-20-i-1	2152.0	0.00	0.82	60-20-i-1	2150.0	0.00	1.37	60-20-i-1	2150.0	0.00	0.74
60-20-i-2	2134.0	0.00	0.78	60-20-i-2	2130.0	0.00	7.09	60-20-i-2	2129.0	0.00	0.65
60-20-i-3	2028.0	0.00	3.19	60-20-i-3	2026.0	0.00	8.57	60-20-i-3	2025.0	0.00	0.73
60-20-i-4	-	-	-	60-20-i-4	-	-	-	60-20-i-4	2026.0	0.00	0.74
60-20-i-5	-	-	-	60-20-i-5	2150.0	0.00	0.71	60-20-i-5	2150.0	0.00	0.63
60-30-i-1	-	-	-	60-30-i-1	3130.0	0.00	6.84	60-30-i-1	3128.0	0.00	1.30
60-30-i-2	-	-	-	60-30-i-2	-	-	-	60-30-i-2	3204.0	0.00	0.59
60-30-i-3	-	-	-	60-30-i-3	3185.0	0.00	1.34	60-30-i-3	3184.0	0.00	1.14
60-30-i-4	-	-	-	60-30-i-4	3183.0	0.00	6.84	60-30-i-4	3181.0	0.00	6.76
60-30-i-5	-	-	-	60-30-i-5	-	-	-	60-30-i-5	3191.0	0.00	1.26
90-10-i-1	-	-	-	90-10-i-1	1636.0	0.00	0.75	90-10-i-1	1636.0	0.00	7.17
90-10-i-2	-	-	-	90-10-i-2	1677.0	0.00	1.37	90-10-i-2	1677.0	0.00	1.35
90-10-i-3	1626.0	0.00	8.52	90-10-i-3	1626.0	0.00	9.43	90-10-i-3	1624.0	0.00	0.75
90-10-i-4	-	-	-	90-10-i-4	-	-	-	90-10-i-4	1649.0	0.00	0.73
90-10-i-5	-	-	-	90-10-i-5	-	-	-	90-10-i-5	1644.0	0.00	0.81
90-20-i-1	-	-	-	90-20-i-1	3307.0	0.00	1.22	90-20-i-1	3307.0	0.00	1.17
90-20-i-2	-	-	-	90-20-i-2	3329.0	0.00	1.79	90-20-i-2	3327.0	0.00	1.12
90-20-i-3	-	-	-	90-20-i-3	3330.0	0.00	4.09	90-20-i-3	3327.0	0.00	1.13
90-20-i-4	3322.0	0.00	0.78	90-20-i-4	3317.0	0.00	2.37	90-20-i-4	3315.0	0.00	0.69
90-20-i-5	-	-	-	90-20-i-5	3288.0	0.00	0.72	90-20-i-5	3286.0	0.00	3.35
90-30-i-1	-	-	-	90-30-i-1	4948.0	0.00	0.81	90-30-i-1	4948.0	0.00	0.64
90-30-i-2	-	-	-	90-30-i-2	-	-	-	90-30-i-2	4835.0	0.00	6.66
90-30-i-3	-	-	-	90-30-i-3	-	-	-	90-30-i-3	5012.0	0.00	0.66
90-30-i-4	-	-	-	90-30-i-4	4720.0	0.00	1.13	90-30-i-4	4720.0	0.00	1.10
90-30-i-5	-	-	-	90-30-i-5	4928.0	0.00	2.84	90-30-i-5	4928.0	0.00	0.69
120-10-i-1	2214.0	0.00	1.45	120-10-i-1	2213.0	0.00	8.62	120-10-i-1	2209.0	0.00	0.68
120-10-i-2	-	-	-	120-10-i-2	-	-	-	120-10-i-2	2216.0	0.00	0.68
120-10-i-3	-	-	-	120-10-i-3	-	-	-	120-10-i-3	2223.0	0.00	0.71
120-10-i-4	-	-	-	120-10-i-4	2176.0	0.00	0.68	120-10-i-4	2176.0	0.00	0.72
120-10-i-5	-	-	-	120-10-i-5	-	-	-	120-10-i-5	2253.0	0.00	7.89
120-20-i-1	-	-	-	120-20-i-1	-	-	-	120-20-i-1	4583.0	0.00	7.13
120-20-i-2	-	-	-	120-20-i-2	-	-	-	120-20-i-2	4541.0	0.00	0.64
120-20-i-3	-	-	-	120-20-i-3	4520.0	0.00	6.24	120-20-i-3	4520.0	0.00	0.61
120-20-i-4	-	-	-	120-20-i-4	-	-	-	120-20-i-4	4431.0	0.00	1.80
120-20-i-5	4486.0	0.00	5.28	120-20-i-5	4485.0	0.00	7.44	120-20-i-5	4485.0	0.00	8.16
120-30-i-1	6724.0	0.00	3.73	120-30-i-1	6720.0	0.00	1.86	120-30-i-1	6717.0	0.00	1.68
120-30-i-2	6786.0	0.00	0.93	120-30-i-2	6784.0	0.00	0.75	120-30-i-2	6784.0	0.00	1.24
120-30-i-3	6685.0	0.00	0.88	120-30-i-3	6685.0	0.00	0.87	120-30-i-3	6685.0	0.00	0.74
120-30-i-4	-	-	-	120-30-i-4	6703.0	0.00	0.85	120-30-i-4	6700.0	0.00	1.20
120-30-i-5	-	-	-	120-30-i-5	-	-	-	120-30-i-5	6635.0	0.00	1.74

1.2. Results on the instances with sf execution mode

Table 4 Detailed Results of Ha26 on Uniform-Weight Instances in S_A with sf execution mode ($ws = 0.4 - 0.6$)

ws-0.4				ws-0.5				ws-0.6			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-sf-1	1044.0	0.00	4.10	60-10-sf-1	1036.0	0.00	9.43	60-10-sf-1	1034.0	0.00	6.48
60-10-sf-2	-	-	-	60-10-sf-2	-	-	-	60-10-sf-2	1052.0	0.00	7.28
60-10-sf-3	1075.0	0.00	7.22	60-10-sf-3	1070.0	0.00	7.42	60-10-sf-3	1069.0	0.00	8.33
60-10-sf-4	-	-	-	60-10-sf-4	1063.0	0.00	0.89	60-10-sf-4	1057.0	0.00	7.45
60-10-sf-5	-	-	-	60-10-sf-5	1063.0	0.00	1.88	60-10-sf-5	1058.0	0.00	11.72
60-20-sf-1	-	-	-	60-20-sf-1	2142.0	0.00	0.90	60-20-sf-1	2137.0	0.00	7.20
60-20-sf-2	-	-	-	60-20-sf-2	-	-	-	60-20-sf-2	2126.0	0.00	9.04
60-20-sf-3	-	-	-	60-20-sf-3	-	-	-	60-20-sf-3	-	-	-
60-20-sf-4	-	-	-	60-20-sf-4	2018.0	0.00	7.63	60-20-sf-4	2011.0	0.00	3.33
60-20-sf-5	-	-	-	60-20-sf-5	2147.0	0.00	8.74	60-20-sf-5	2139.0	0.00	8.91
60-30-sf-1	-	-	-	60-30-sf-1	3108.0	0.00	2.58	60-30-sf-1	3099.0	0.00	8.20
60-30-sf-2	-	-	-	60-30-sf-2	-	-	-	60-30-sf-2	3183.0	0.00	149.66
60-30-sf-3	-	-	-	60-30-sf-3	-	-	-	60-30-sf-3	3171.0	0.00	1.68
60-30-sf-4	-	-	-	60-30-sf-4	3177.0	0.00	11.25	60-30-sf-4	3162.0	0.00	69.22
60-30-sf-5	-	-	-	60-30-sf-5	-	-	-	60-30-sf-5	3172.0	0.00	10.55
90-10-sf-1	-	-	-	90-10-sf-1	-	-	-	90-10-sf-1	1623.0	0.00	7.21
90-10-sf-2	-	-	-	90-10-sf-2	-	-	-	90-10-sf-2	1666.0	0.00	0.84
90-10-sf-3	-	-	-	90-10-sf-3	1614.0	0.00	1.62	90-10-sf-3	1611.0	0.00	0.82
90-10-sf-4	-	-	-	90-10-sf-4	-	-	-	90-10-sf-4	-	-	-
90-10-sf-5	1642.0	0.00	8.51	90-10-sf-5	1636.0	0.00	8.34	90-10-sf-5	1632.0	0.00	3.00
90-20-sf-1	-	-	-	90-20-sf-1	3291.0	0.00	12.31	90-20-sf-1	3281.0	0.00	8.74
90-20-sf-2	-	-	-	90-20-sf-2	3305.0	0.00	13.57	90-20-sf-2	3295.0	0.00	7.19
90-20-sf-3	-	-	-	90-20-sf-3	3308.0	0.00	10.26	90-20-sf-3	3298.0	0.00	7.63
90-20-sf-4	-	-	-	90-20-sf-4	-	-	-	90-20-sf-4	3301.0	0.00	15.20
90-20-sf-5	-	-	-	90-20-sf-5	-	-	-	90-20-sf-5	3262.0	0.00	43.98
90-30-sf-1	-	-	-	90-30-sf-1	-	-	-	90-30-sf-1	-	-	-
90-30-sf-2	-	-	-	90-30-sf-2	-	-	-	90-30-sf-2	-	-	-
90-30-sf-3	-	-	-	90-30-sf-3	4989.0	0.00	182.35	90-30-sf-3	4977.0	0.00	110.06
90-30-sf-4	-	-	-	90-30-sf-4	4705.0	0.00	19.02	90-30-sf-4	4683.0	0.00	32.65
90-30-sf-5	-	-	-	90-30-sf-5	4903.0	0.00	180.74	90-30-sf-5	4891.0	0.00	41.12
120-10-sf-1	-	-	-	120-10-sf-1	-	-	-	120-10-sf-1	2197.0	0.00	7.61
120-10-sf-2	-	-	-	120-10-sf-2	-	-	-	120-10-sf-2	2206.0	0.00	9.33
120-10-sf-3	-	-	-	120-10-sf-3	-	-	-	120-10-sf-3	2208.0	0.00	2.84
120-10-sf-4	-	-	-	120-10-sf-4	2158.0	0.00	7.53	120-10-sf-4	2157.0	0.00	7.22
120-10-sf-5	-	-	-	120-10-sf-5	-	-	-	120-10-sf-5	2239.0	0.00	1.27
120-20-sf-1	4558.0	0.00	12.41	120-20-sf-1	4543.0	0.00	3.28	120-20-sf-1	4538.0	0.00	6.72
120-20-sf-2	-	-	-	120-20-sf-2	4520.0	0.00	9.64	120-20-sf-2	4506.0	0.00	9.56
120-20-sf-3	-	-	-	120-20-sf-3	-	-	-	120-20-sf-3	4479.0	0.00	136.46
120-20-sf-4	-	-	-	120-20-sf-4	-	-	-	120-20-sf-4	-	-	-
120-20-sf-5	-	-	-	120-20-sf-5	-	-	-	120-20-sf-5	-	-	-
120-30-sf-1	-	-	-	120-30-sf-1	6686.0	3.05	182.77	120-30-sf-1	6661.0	0.00	10.63
120-30-sf-2	-	-	-	120-30-sf-2	6751.0	0.00	183.45	120-30-sf-2	6734.0	0.00	154.32
120-30-sf-3	-	-	-	120-30-sf-3	6647.0	0.00	22.78	120-30-sf-3	6636.0	0.00	7.92
120-30-sf-4	-	-	-	120-30-sf-4	6676.0	0.00	182.45	120-30-sf-4	6654.0	0.00	25.87
120-30-sf-5	-	-	-	120-30-sf-5	-	-	-	120-30-sf-5	-	-	-

Table 5 Detailed Results of Ha26 on Uniform-Weight Instances in S_A with sf execution mode ($ws = 0.7 - 0.9$)

ws-0.7				ws-0.8				ws-0.9			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-sf-1	1031.0	0.00	0.96	60-10-sf-1	1031.0	0.00	10.89	60-10-sf-1	1031.0	0.00	0.79
60-10-sf-2	1049.0	0.00	10.92	60-10-sf-2	1049.0	0.00	10.91	60-10-sf-2	1048.0	0.00	10.23
60-10-sf-3	1067.0	0.00	3.03	60-10-sf-3	1063.0	0.00	10.07	60-10-sf-3	1063.0	0.00	11.00
60-10-sf-4	1053.0	0.00	1.74	60-10-sf-4	1052.0	0.00	2.42	60-10-sf-4	1051.0	0.00	0.83
60-10-sf-5	1056.0	0.00	11.12	60-10-sf-5	1055.0	0.00	3.32	60-10-sf-5	1053.0	0.00	0.89
60-20-sf-1	2134.0	0.00	7.53	60-20-sf-1	2132.0	0.00	0.88	60-20-sf-1	2132.0	0.00	1.33
60-20-sf-2	2116.0	0.00	2.15	60-20-sf-2	2110.0	0.00	9.94	60-20-sf-2	2109.0	0.00	1.43
60-20-sf-3	2018.0	0.00	3.82	60-20-sf-3	2017.0	0.00	8.22	60-20-sf-3	2014.0	0.00	0.76
60-20-sf-4	2009.0	0.00	7.29	60-20-sf-4	2007.0	0.00	1.45	60-20-sf-4	2006.0	0.00	1.98
60-20-sf-5	2134.0	0.00	2.55	60-20-sf-5	2133.0	0.00	1.91	60-20-sf-5	2132.0	0.00	0.74
60-30-sf-1	3096.0	0.00	6.35	60-30-sf-1	3093.0	0.00	5.66	60-30-sf-1	3092.0	0.00	2.86
60-30-sf-2	3169.0	0.00	8.47	60-30-sf-2	3166.0	0.00	4.23	60-30-sf-2	3165.0	0.00	0.71
60-30-sf-3	3162.0	0.00	2.64	60-30-sf-3	3157.0	0.00	5.53	60-30-sf-3	3155.0	0.00	0.71
60-30-sf-4	3156.0	0.00	15.25	60-30-sf-4	3152.0	0.00	7.31	60-30-sf-4	3150.0	0.00	8.10
60-30-sf-5	3164.0	0.00	9.12	60-30-sf-5	3162.0	0.00	7.55	60-30-sf-5	3160.0	0.00	7.77
90-10-sf-1	1622.0	0.00	2.48	90-10-sf-1	1619.0	0.00	1.97	90-10-sf-1	1619.0	0.00	3.97
90-10-sf-2	1662.0	0.00	6.92	90-10-sf-2	1659.0	0.00	1.96	90-10-sf-2	1658.0	0.00	8.17
90-10-sf-3	1610.0	0.00	0.83	90-10-sf-3	1610.0	0.00	0.93	90-10-sf-3	1610.0	0.00	0.84
90-10-sf-4	-	-	-	90-10-sf-4	1635.0	0.00	0.76	90-10-sf-4	1635.0	0.00	9.15
90-10-sf-5	1630.0	0.00	2.27	90-10-sf-5	1629.0	0.00	0.78	90-10-sf-5	1629.0	0.00	9.17
90-20-sf-1	3275.0	0.00	4.84	90-20-sf-1	3273.0	0.00	1.31	90-20-sf-1	3273.0	0.00	7.42
90-20-sf-2	3293.0	0.00	8.01	90-20-sf-2	3291.0	0.00	7.18	90-20-sf-2	3289.0	0.00	1.28
90-20-sf-3	3296.0	0.00	6.92	90-20-sf-3	3295.0	0.00	4.27	90-20-sf-3	3295.0	0.00	5.34
90-20-sf-4	3292.0	0.00	7.31	90-20-sf-4	3288.0	0.00	8.29	90-20-sf-4	3286.0	0.00	7.64
90-20-sf-5	3257.0	0.00	7.05	90-20-sf-5	3251.0	0.00	2.11	90-20-sf-5	3249.0	0.00	1.30
90-30-sf-1	4903.0	0.00	16.82	90-30-sf-1	4891.0	0.00	7.46	90-30-sf-1	4889.0	0.00	1.89
90-30-sf-2	4801.0	0.00	13.32	90-30-sf-2	4795.0	0.00	7.90	90-30-sf-2	4791.0	0.00	1.33
90-30-sf-3	4970.0	0.00	15.66	90-30-sf-3	4969.0	0.00	11.85	90-30-sf-3	4966.0	0.00	1.60
90-30-sf-4	4675.0	0.00	10.47	90-30-sf-4	4673.0	0.00	1.47	90-30-sf-4	4672.0	0.00	0.68
90-30-sf-5	4886.0	0.00	5.01	90-30-sf-5	4885.0	0.00	1.93	90-30-sf-5	4884.0	0.00	0.74
120-10-sf-1	2194.0	0.00	7.38	120-10-sf-1	2192.0	0.00	0.67	120-10-sf-1	2192.0	0.00	8.90
120-10-sf-2	2206.0	0.00	1.45	120-10-sf-2	2204.0	0.00	1.38	120-10-sf-2	2204.0	0.00	1.40
120-10-sf-3	2206.0	0.00	4.09	120-10-sf-3	2205.0	0.00	8.82	120-10-sf-3	2205.0	0.00	2.21
120-10-sf-4	2156.0	0.00	7.23	120-10-sf-4	2155.0	0.00	6.62	120-10-sf-4	2150.0	0.00	0.68
120-10-sf-5	2237.0	0.00	7.98	120-10-sf-5	2235.0	0.00	3.60	120-10-sf-5	2235.0	0.00	3.58
120-20-sf-1	4536.0	0.00	2.62	120-20-sf-1	4533.0	0.00	0.76	120-20-sf-1	4533.0	0.00	7.13
120-20-sf-2	4498.0	0.00	4.73	120-20-sf-2	4495.0	0.00	5.98	120-20-sf-2	4492.0	0.00	3.23
120-20-sf-3	4470.0	0.00	103.08	120-20-sf-3	4464.0	0.00	6.98	120-20-sf-3	4461.0	0.00	1.22
120-20-sf-4	-	-	-	120-20-sf-4	4386.0	0.00	2.03	120-20-sf-4	4385.0	0.00	6.98
120-20-sf-5	-	-	-	120-20-sf-5	4446.0	0.00	0.83	120-20-sf-5	4446.0	0.00	0.72
120-30-sf-1	6656.0	0.00	7.97	120-30-sf-1	6655.0	0.00	1.31	120-30-sf-1	6655.0	0.00	1.27
120-30-sf-2	6727.0	0.00	26.85	120-30-sf-2	6724.0	0.00	3.23	120-30-sf-2	6722.0	0.00	0.87
120-30-sf-3	6633.0	0.00	1.08	120-30-sf-3	6633.0	0.00	6.95	120-30-sf-3	6633.0	0.00	1.37
120-30-sf-4	6640.0	0.00	5.12	120-30-sf-4	6639.0	0.00	2.83	120-30-sf-4	6636.0	0.00	0.84
120-30-sf-5	6568.0	0.00	11.99	120-30-sf-5	6565.0	0.00	4.12	120-30-sf-5	6558.0	0.00	3.26

1.3. Results on the instances with sif execution mode

Table 6 Detailed Results of Ha26 on Uniform-Weight Instances in S_A with sif execution mode ($ws = 0.4 - 0.6$)

ws-0.4				ws-0.5				ws-0.6			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-sif-1	1042.0	0.00	12.55	60-10-sif-1	1038.0	0.00	5.48	60-10-sif-1	1035.0	0.00	9.14
60-10-sif-2	-	-	-	60-10-sif-2	1063.0	0.00	1.63	60-10-sif-2	1054.0	0.00	0.87
60-10-sif-3	1076.0	0.00	9.57	60-10-sif-3	1071.0	0.00	8.49	60-10-sif-3	1069.0	0.00	7.10
60-10-sif-4	-	-	-	60-10-sif-4	1063.0	0.00	0.90	60-10-sif-4	1057.0	0.00	8.21
60-10-sif-5	1062.0	0.00	0.93	60-10-sif-5	1059.0	0.00	2.67	60-10-sif-5	1057.0	0.00	4.48
60-20-sif-1	-	-	-	60-20-sif-1	2142.0	0.00	0.98	60-20-sif-1	2137.0	0.00	8.65
60-20-sif-2	-	-	-	60-20-sif-2	2126.0	0.00	11.88	60-20-sif-2	2120.0	0.00	12.50
60-20-sif-3	-	-	-	60-20-sif-3	-	-	-	60-20-sif-3	2024.0	0.00	6.05
60-20-sif-4	-	-	-	60-20-sif-4	2017.0	0.00	5.68	60-20-sif-4	2011.0	0.00	5.32
60-20-sif-5	-	-	-	60-20-sif-5	2146.0	0.00	8.60	60-20-sif-5	2139.0	0.00	9.15
60-30-sif-1	-	-	-	60-30-sif-1	3105.0	0.00	14.09	60-30-sif-1	3101.0	0.00	9.49
60-30-sif-2	-	-	-	60-30-sif-2	-	-	-	60-30-sif-2	3184.0	0.00	181.79
60-30-sif-3	-	-	-	60-30-sif-3	-	-	-	60-30-sif-3	3166.0	0.00	9.50
60-30-sif-4	-	-	-	60-30-sif-4	-	-	-	60-30-sif-4	3161.0	0.00	4.09
60-30-sif-5	-	-	-	60-30-sif-5	-	-	-	60-30-sif-5	3175.0	0.00	25.47
90-10-sif-1	-	-	-	90-10-sif-1	1629.0	0.00	18.18	90-10-sif-1	1624.0	0.00	7.65
90-10-sif-2	-	-	-	90-10-sif-2	1673.0	0.00	5.94	90-10-sif-2	1663.0	0.00	4.65
90-10-sif-3	-	-	-	90-10-sif-3	1617.0	0.00	9.49	90-10-sif-3	1612.0	0.00	0.84
90-10-sif-4	-	-	-	90-10-sif-4	-	-	-	90-10-sif-4	-	-	-
90-10-sif-5	-	-	-	90-10-sif-5	1636.0	0.00	8.80	90-10-sif-5	1632.0	0.00	3.07
90-20-sif-1	-	-	-	90-20-sif-1	3295.0	0.00	8.09	90-20-sif-1	3281.0	0.00	11.67
90-20-sif-2	-	-	-	90-20-sif-2	3315.0	0.00	103.87	90-20-sif-2	3300.0	0.00	56.19
90-20-sif-3	-	-	-	90-20-sif-3	3310.0	0.00	30.53	90-20-sif-3	3299.0	0.00	9.80
90-20-sif-4	-	-	-	90-20-sif-4	-	-	-	90-20-sif-4	3300.0	0.00	11.95
90-20-sif-5	-	-	-	90-20-sif-5	-	-	-	90-20-sif-5	3262.0	0.00	9.74
90-30-sif-1	-	-	-	90-30-sif-1	4934.0	4.87	185.23	90-30-sif-1	4912.0	2.94	182.04
90-30-sif-2	-	-	-	90-30-sif-2	-	-	-	90-30-sif-2	4810.0	0.00	20.84
90-30-sif-3	-	-	-	90-30-sif-3	4991.0	0.00	181.53	90-30-sif-3	4978.0	0.00	47.02
90-30-sif-4	-	-	-	90-30-sif-4	4698.0	2.16	181.68	90-30-sif-4	4681.0	0.00	39.88
90-30-sif-5	4932.0	0.00	5.06	90-30-sif-5	4902.0	0.00	181.47	90-30-sif-5	4890.0	0.00	44.66
120-10-sif-1	-	-	-	120-10-sif-1	-	-	-	120-10-sif-1	2200.0	0.00	8.05
120-10-sif-2	2214.0	0.00	7.33	120-10-sif-2	2207.0	0.00	0.82	120-10-sif-2	2206.0	0.00	0.72
120-10-sif-3	-	-	-	120-10-sif-3	-	-	-	120-10-sif-3	2209.0	0.00	5.96
120-10-sif-4	-	-	-	120-10-sif-4	2159.0	0.00	8.08	120-10-sif-4	2155.0	0.00	3.69
120-10-sif-5	-	-	-	120-10-sif-5	-	-	-	120-10-sif-5	2240.0	0.00	5.85
120-20-sif-1	-	-	-	120-20-sif-1	4545.0	0.00	9.37	120-20-sif-1	4538.0	0.00	8.39
120-20-sif-2	-	-	-	120-20-sif-2	4519.0	0.00	10.76	120-20-sif-2	4503.0	0.00	7.43
120-20-sif-3	-	-	-	120-20-sif-3	-	-	-	120-20-sif-3	4482.0	0.00	73.10
120-20-sif-4	-	-	-	120-20-sif-4	4407.0	0.00	27.63	120-20-sif-4	4397.0	0.00	35.80
120-20-sif-5	-	-	-	120-20-sif-5	-	-	-	120-20-sif-5	4457.0	0.00	27.02
120-30-sif-1	-	-	-	120-30-sif-1	6706.0	6.71	182.48	120-30-sif-1	6666.0	0.00	182.61
120-30-sif-2	-	-	-	120-30-sif-2	6749.0	2.22	187.05	120-30-sif-2	6733.0	0.00	180.76
120-30-sif-3	-	-	-	120-30-sif-3	6656.0	0.00	142.31	120-30-sif-3	6636.0	0.00	5.08
120-30-sif-4	-	-	-	120-30-sif-4	6684.0	3.10	182.21	120-30-sif-4	6652.0	0.00	182.22
120-30-sif-5	-	-	-	120-30-sif-5	-	-	-	120-30-sif-5	6579.0	0.00	183.37

Table 7 Detailed Results of Ha26 on Uniform-Weight Instances in S_A with sif execution mode ($ws = 0.7 - 0.9$)

ws-0.7				ws-0.8				ws-0.9			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-sif-1	1033.0	0.00	5.04	60-10-sif-1	1031.0	0.00	10.79	60-10-sif-1	1031.0	0.00	0.79
60-10-sif-2	1053.0	0.00	8.76	60-10-sif-2	1049.0	0.00	10.52	60-10-sif-2	1048.0	0.00	10.35
60-10-sif-3	1068.0	0.00	8.12	60-10-sif-3	1066.0	0.00	11.20	60-10-sif-3	1063.0	0.00	10.96
60-10-sif-4	1053.0	0.00	5.25	60-10-sif-4	1052.0	0.00	2.45	60-10-sif-4	1051.0	0.00	0.84
60-10-sif-5	1055.0	0.00	0.76	60-10-sif-5	1054.0	0.00	1.56	60-10-sif-5	1053.0	0.00	0.89
60-20-sif-1	2134.0	0.00	7.66	60-20-sif-1	2132.0	0.00	0.88	60-20-sif-1	2132.0	0.00	1.38
60-20-sif-2	2115.0	0.00	4.73	60-20-sif-2	2113.0	0.00	9.28	60-20-sif-2	2109.0	0.00	2.12
60-20-sif-3	2016.0	0.00	0.90	60-20-sif-3	2016.0	0.00	1.66	60-20-sif-3	2014.0	0.00	0.79
60-20-sif-4	2010.0	0.00	11.55	60-20-sif-4	2007.0	0.00	1.50	60-20-sif-4	2006.0	0.00	2.02
60-20-sif-5	2134.0	0.00	8.14	60-20-sif-5	2133.0	0.00	1.88	60-20-sif-5	2132.0	0.00	0.67
60-30-sif-1	3096.0	0.00	12.66	60-30-sif-1	3091.0	0.00	1.30	60-30-sif-1	3090.0	0.00	1.99
60-30-sif-2	3171.0	0.00	56.84	60-30-sif-2	3166.0	0.00	3.31	60-30-sif-2	3165.0	0.00	0.67
60-30-sif-3	3161.0	0.00	6.66	60-30-sif-3	3158.0	0.00	1.48	60-30-sif-3	3155.0	0.00	1.31
60-30-sif-4	3155.0	0.00	20.02	60-30-sif-4	3152.0	0.00	9.21	60-30-sif-4	3150.0	0.00	0.69
60-30-sif-5	3165.0	0.00	10.85	60-30-sif-5	3162.0	0.00	7.10	60-30-sif-5	3160.0	0.00	0.74
90-10-sif-1	1621.0	0.00	7.60	90-10-sif-1	1620.0	0.00	4.18	90-10-sif-1	1617.0	0.00	9.46
90-10-sif-2	1660.0	0.00	8.90	90-10-sif-2	1659.0	0.00	7.74	90-10-sif-2	1658.0	0.00	8.21
90-10-sif-3	1610.0	0.00	0.80	90-10-sif-3	1610.0	0.00	0.94	90-10-sif-3	1610.0	0.00	0.85
90-10-sif-4	1637.0	0.00	2.13	90-10-sif-4	1637.0	0.00	3.53	90-10-sif-4	1635.0	0.00	9.42
90-10-sif-5	1631.0	0.00	9.57	90-10-sif-5	1629.0	0.00	0.78	90-10-sif-5	1629.0	0.00	9.31
90-20-sif-1	3275.0	0.00	2.73	90-20-sif-1	3273.0	0.00	1.34	90-20-sif-1	3273.0	0.00	2.54
90-20-sif-2	3293.0	0.00	6.48	90-20-sif-2	3291.0	0.00	5.77	90-20-sif-2	3289.0	0.00	1.29
90-20-sif-3	3296.0	0.00	7.12	90-20-sif-3	3293.0	0.00	1.89	90-20-sif-3	3293.0	0.00	2.43
90-20-sif-4	3292.0	0.00	8.53	90-20-sif-4	3288.0	0.00	7.93	90-20-sif-4	3286.0	0.00	3.23
90-20-sif-5	3255.0	0.00	8.32	90-20-sif-5	3251.0	0.00	9.32	90-20-sif-5	3249.0	0.00	1.46
90-30-sif-1	4900.0	0.00	28.90	90-30-sif-1	4898.0	0.00	6.96	90-30-sif-1	4889.0	0.00	7.81
90-30-sif-2	4798.0	0.00	8.38	90-30-sif-2	4795.0	0.00	7.34	90-30-sif-2	4793.0	0.00	3.08
90-30-sif-3	4972.0	0.00	106.23	90-30-sif-3	4969.0	0.00	5.09	90-30-sif-3	4966.0	0.00	1.65
90-30-sif-4	4675.0	0.00	5.08	90-30-sif-4	4673.0	0.00	1.53	90-30-sif-4	4672.0	0.00	1.41
90-30-sif-5	4886.0	0.00	3.86	90-30-sif-5	4885.0	0.00	3.15	90-30-sif-5	4884.0	0.00	0.91
120-10-sif-1	2198.0	0.00	7.56	120-10-sif-1	2192.0	0.00	0.77	120-10-sif-1	2192.0	0.00	1.39
120-10-sif-2	2205.0	0.00	1.45	120-10-sif-2	2203.0	0.00	0.74	120-10-sif-2	2203.0	0.00	0.74
120-10-sif-3	2205.0	0.00	0.81	120-10-sif-3	2204.0	0.00	0.81	120-10-sif-3	2204.0	0.00	1.43
120-10-sif-4	2155.0	0.00	7.03	120-10-sif-4	2154.0	0.00	2.07	120-10-sif-4	2150.0	0.00	0.69
120-10-sif-5	2236.0	0.00	2.90	120-10-sif-5	2234.0	0.00	1.85	120-10-sif-5	2234.0	0.00	7.53
120-20-sif-1	4536.0	0.00	1.50	120-20-sif-1	4535.0	0.00	1.87	120-20-sif-1	4533.0	0.00	0.81
120-20-sif-2	4498.0	0.00	7.07	120-20-sif-2	4495.0	0.00	0.81	120-20-sif-2	4492.0	0.00	4.71
120-20-sif-3	4473.0	0.00	68.69	120-20-sif-3	4464.0	0.00	2.94	120-20-sif-3	4461.0	0.00	1.20
120-20-sif-4	4393.0	0.00	64.15	120-20-sif-4	4392.0	0.00	2.71	120-20-sif-4	4385.0	0.00	7.17
120-20-sif-5	4450.0	0.00	27.14	120-20-sif-5	4446.0	0.00	1.46	120-20-sif-5	4446.0	0.00	0.71
120-30-sif-1	6658.0	0.00	99.03	120-30-sif-1	6655.0	0.00	1.01	120-30-sif-1	6655.0	0.00	0.94
120-30-sif-2	6726.0	0.00	59.09	120-30-sif-2	6724.0	0.00	2.25	120-30-sif-2	6722.0	0.00	0.89
120-30-sif-3	6634.0	0.00	8.05	120-30-sif-3	6633.0	0.00	1.48	120-30-sif-3	6633.0	0.00	1.96
120-30-sif-4	6640.0	0.00	31.37	120-30-sif-4	6639.0	0.00	16.90	120-30-sif-4	6639.0	0.00	3.31
120-30-sif-5	6566.0	0.00	37.90	120-30-sif-5	6563.0	0.00	8.16	120-30-sif-5	6563.0	0.00	2.59

2. Detailed Results of Standard-Weight Instances on S_A under WTFRPA Setting

We first report the results of Ha26 on the instance with $ws = 0.3$. Among these instances, only the instance "60-10-sif-0.3-3" is feasible.

Table 8 Detailed Results of Ha26 on 60-10-sif-0.3-3

ws-0.3			
Name	Cost	Gap‰	CPU
60-10-sif-3	2301.0	0.00	1.21

2.1. Results on the instances with i execution mode

Table 9 Detailed Results of Ha26 on Standard-Weight Instances in S_A with i execution mode ($ws = 0.4 - 0.6$)

ws-0.4				ws-0.5				ws-0.6			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-i-1	-	-	-	60-10-i-1	-	-	-	60-10-i-1	-	-	-
60-10-i-2	-	-	-	60-10-i-2	-	-	-	60-10-i-2	-	-	-
60-10-i-3	-	-	-	60-10-i-3	-	-	-	60-10-i-3	2303.0	0.00	11.05
60-10-i-4	-	-	-	60-10-i-4	-	-	-	60-10-i-4	-	-	-
60-10-i-5	-	-	-	60-10-i-5	-	-	-	60-10-i-5	-	-	-
60-20-i-1	-	-	-	60-20-i-1	-	-	-	60-20-i-1	-	-	-
60-20-i-2	-	-	-	60-20-i-2	-	-	-	60-20-i-2	-	-	-
60-20-i-3	-	-	-	60-20-i-3	-	-	-	60-20-i-3	-	-	-
60-20-i-4	-	-	-	60-20-i-4	-	-	-	60-20-i-4	-	-	-
60-20-i-5	-	-	-	60-20-i-5	-	-	-	60-20-i-5	-	-	-
60-30-i-1	-	-	-	60-30-i-1	-	-	-	60-30-i-1	-	-	-
60-30-i-2	-	-	-	60-30-i-2	-	-	-	60-30-i-2	-	-	-
60-30-i-3	-	-	-	60-30-i-3	-	-	-	60-30-i-3	-	-	-
60-30-i-4	-	-	-	60-30-i-4	-	-	-	60-30-i-4	-	-	-
60-30-i-5	-	-	-	60-30-i-5	-	-	-	60-30-i-5	-	-	-
90-10-i-1	-	-	-	90-10-i-1	-	-	-	90-10-i-1	-	-	-
90-10-i-2	-	-	-	90-10-i-2	-	-	-	90-10-i-2	-	-	-
90-10-i-3	-	-	-	90-10-i-3	-	-	-	90-10-i-3	-	-	-
90-10-i-4	-	-	-	90-10-i-4	-	-	-	90-10-i-4	-	-	-
90-10-i-5	-	-	-	90-10-i-5	-	-	-	90-10-i-5	-	-	-
90-20-i-1	-	-	-	90-20-i-1	-	-	-	90-20-i-1	-	-	-
90-20-i-2	-	-	-	90-20-i-2	-	-	-	90-20-i-2	-	-	-
90-20-i-3	-	-	-	90-20-i-3	-	-	-	90-20-i-3	-	-	-
90-20-i-4	-	-	-	90-20-i-4	-	-	-	90-20-i-4	-	-	-
90-20-i-5	-	-	-	90-20-i-5	-	-	-	90-20-i-5	-	-	-
90-30-i-1	-	-	-	90-30-i-1	-	-	-	90-30-i-1	-	-	-
90-30-i-2	-	-	-	90-30-i-2	-	-	-	90-30-i-2	-	-	-
90-30-i-3	-	-	-	90-30-i-3	-	-	-	90-30-i-3	-	-	-
90-30-i-4	-	-	-	90-30-i-4	-	-	-	90-30-i-4	-	-	-
90-30-i-5	-	-	-	90-30-i-5	-	-	-	90-30-i-5	-	-	-
120-10-i-1	-	-	-	120-10-i-1	-	-	-	120-10-i-1	-	-	-
120-10-i-2	-	-	-	120-10-i-2	-	-	-	120-10-i-2	-	-	-
120-10-i-3	-	-	-	120-10-i-3	-	-	-	120-10-i-3	-	-	-
120-10-i-4	-	-	-	120-10-i-4	-	-	-	120-10-i-4	-	-	-
120-10-i-5	-	-	-	120-10-i-5	-	-	-	120-10-i-5	-	-	-
120-20-i-1	-	-	-	120-20-i-1	-	-	-	120-20-i-1	-	-	-
120-20-i-2	-	-	-	120-20-i-2	-	-	-	120-20-i-2	-	-	-
120-20-i-3	-	-	-	120-20-i-3	-	-	-	120-20-i-3	-	-	-
120-20-i-4	-	-	-	120-20-i-4	-	-	-	120-20-i-4	-	-	-
120-20-i-5	-	-	-	120-20-i-5	-	-	-	120-20-i-5	-	-	-
120-30-i-1	-	-	-	120-30-i-1	-	-	-	120-30-i-1	-	-	-
120-30-i-2	-	-	-	120-30-i-2	-	-	-	120-30-i-2	12026.0	0.00	9.75
120-30-i-3	-	-	-	120-30-i-3	-	-	-	120-30-i-3	11254.0	0.00	1.14
120-30-i-4	-	-	-	120-30-i-4	-	-	-	120-30-i-4	-	-	-
120-30-i-5	-	-	-	120-30-i-5	-	-	-	120-30-i-5	-	-	-

Table 10 Detailed Results of Ha26 on Standard-Weight Instances in S_A with i execution mode ($ws = 0.7 - 0.9$)

ws-0.7				ws-0.8				ws-0.9			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-i-1	-	-	-	60-10-i-1	1881.0	0.00	9.40	60-10-i-1	1879.0	0.00	0.80
60-10-i-2	2019.0	0.00	9.30	60-10-i-2	2016.0	0.00	9.89	60-10-i-2	2011.0	0.00	0.80
60-10-i-3	2289.0	0.00	0.86	60-10-i-3	2289.0	0.00	9.22	60-10-i-3	2289.0	0.00	10.44
60-10-i-4	-	-	-	60-10-i-4	-	-	-	60-10-i-4	-	-	-
60-10-i-5	-	-	-	60-10-i-5	-	-	-	60-10-i-5	1925.0	0.00	9.87
60-20-i-1	3774.0	0.00	0.73	60-20-i-1	3770.0	0.00	1.40	60-20-i-1	3770.0	0.00	1.22
60-20-i-2	3954.0	0.00	0.95	60-20-i-2	3948.0	0.00	7.33	60-20-i-2	3946.0	0.00	0.65
60-20-i-3	3200.0	0.00	8.00	60-20-i-3	3198.0	0.00	8.56	60-20-i-3	3196.0	0.00	0.74
60-20-i-4	-	-	-	60-20-i-4	-	-	-	60-20-i-4	3844.0	0.00	0.73
60-20-i-5	-	-	-	60-20-i-5	3658.0	0.00	0.72	60-20-i-5	3658.0	0.00	0.63
60-30-i-1	-	-	-	60-30-i-1	6732.0	0.00	1.23	60-30-i-1	6725.0	0.00	6.45
60-30-i-2	-	-	-	60-30-i-2	-	-	-	60-30-i-2	5786.0	0.00	0.60
60-30-i-3	-	-	-	60-30-i-3	5340.0	0.00	0.66	60-30-i-3	5338.0	0.00	1.14
60-30-i-4	-	-	-	60-30-i-4	5538.0	0.00	2.38	60-30-i-4	5533.0	0.00	0.71
60-30-i-5	-	-	-	60-30-i-5	-	-	-	60-30-i-5	5328.0	0.00	1.25
90-10-i-1	-	-	-	90-10-i-1	2950.0	0.00	0.76	90-10-i-1	2950.0	0.00	7.18
90-10-i-2	-	-	-	90-10-i-2	3024.0	0.00	1.38	90-10-i-2	3024.0	0.00	1.38
90-10-i-3	2948.0	0.00	8.50	90-10-i-3	2948.0	0.00	9.45	90-10-i-3	2944.0	0.00	0.76
90-10-i-4	-	-	-	90-10-i-4	-	-	-	90-10-i-4	3186.0	0.00	0.73
90-10-i-5	-	-	-	90-10-i-5	-	-	-	90-10-i-5	2641.0	0.00	0.82
90-20-i-1	-	-	-	90-20-i-1	6565.0	0.00	1.27	90-20-i-1	6565.0	0.00	1.16
90-20-i-2	-	-	-	90-20-i-2	6877.0	0.00	2.27	90-20-i-2	6873.0	0.00	6.99
90-20-i-3	-	-	-	90-20-i-3	6184.0	0.00	2.57	90-20-i-3	6175.0	0.00	1.13
90-20-i-4	5311.0	0.00	1.38	90-20-i-4	5303.0	0.00	3.68	90-20-i-4	5301.0	0.00	0.70
90-20-i-5	-	-	-	90-20-i-5	6029.0	0.00	7.08	90-20-i-5	6025.0	0.00	7.07
90-30-i-1	-	-	-	90-30-i-1	9625.0	0.00	0.67	90-30-i-1	9625.0	0.00	0.64
90-30-i-2	-	-	-	90-30-i-2	-	-	-	90-30-i-2	8581.0	0.00	6.60
90-30-i-3	-	-	-	90-30-i-3	-	-	-	90-30-i-3	8719.0	0.00	0.65
90-30-i-4	-	-	-	90-30-i-4	9026.0	0.00	1.16	90-30-i-4	9026.0	0.00	1.10
90-30-i-5	-	-	-	90-30-i-5	8040.0	0.00	1.82	90-30-i-5	8040.0	0.00	0.67
120-10-i-1	3892.0	0.00	2.08	120-10-i-1	3890.0	0.00	4.45	120-10-i-1	3886.0	0.00	0.70
120-10-i-2	-	-	-	120-10-i-2	-	-	-	120-10-i-2	3571.0	0.00	0.70
120-10-i-3	-	-	-	120-10-i-3	-	-	-	120-10-i-3	3440.0	0.00	0.72
120-10-i-4	-	-	-	120-10-i-4	4352.0	0.00	0.70	120-10-i-4	4352.0	0.00	0.73
120-10-i-5	-	-	-	120-10-i-5	-	-	-	120-10-i-5	3858.0	0.00	7.88
120-20-i-1	-	-	-	120-20-i-1	-	-	-	120-20-i-1	8143.0	0.00	7.08
120-20-i-2	-	-	-	120-20-i-2	-	-	-	120-20-i-2	9375.0	0.00	0.65
120-20-i-3	-	-	-	120-20-i-3	9742.0	0.00	0.68	120-20-i-3	9742.0	0.00	0.64
120-20-i-4	-	-	-	120-20-i-4	-	-	-	120-20-i-4	7792.0	0.00	1.80
120-20-i-5	7534.0	0.00	12.41	120-20-i-5	7532.0	0.00	11.37	120-20-i-5	7532.0	0.00	7.05
120-30-i-1	13095.0	0.00	6.71	120-30-i-1	13086.0	0.00	2.47	120-30-i-1	13080.0	0.00	2.17
120-30-i-2	12008.0	0.00	1.51	120-30-i-2	12002.0	0.00	1.17	120-30-i-2	12002.0	0.00	1.21
120-30-i-3	11246.0	0.00	0.83	120-30-i-3	11246.0	0.00	2.65	120-30-i-3	11246.0	0.00	0.72
120-30-i-4	-	-	-	120-30-i-4	11498.0	0.00	0.83	120-30-i-4	11490.0	0.00	1.17
120-30-i-5	-	-	-	120-30-i-5	-	-	-	120-30-i-5	13444.0	0.00	0.74

2.2. Results on the instances with sf execution mode

Table 11 Detailed Results of Ha26 on Standard-Weight Instances in S_A with sf execution mode ($ws = 0.4 - 0.6$)

ws-0.4				ws-0.5				ws-0.6			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-sf-1	1884.0	0.00	9.69	60-10-sf-1	1868.0	0.00	9.26	60-10-sf-1	1864.0	0.00	6.33
60-10-sf-2	-	-	-	60-10-sf-2	-	-	-	60-10-sf-2	1998.0	0.00	5.52
60-10-sf-3	2283.0	0.00	7.06	60-10-sf-3	2269.0	0.00	8.39	60-10-sf-3	2265.0	0.00	2.97
60-10-sf-4	-	-	-	60-10-sf-4	2037.0	0.00	0.89	60-10-sf-4	2019.0	0.00	3.48
60-10-sf-5	-	-	-	60-10-sf-5	1919.0	0.00	1.87	60-10-sf-5	1910.0	0.00	11.75
60-20-sf-1	-	-	-	60-20-sf-1	3757.0	0.00	0.93	60-20-sf-1	3747.0	0.00	8.41
60-20-sf-2	-	-	-	60-20-sf-2	-	-	-	60-20-sf-2	3937.0	0.00	8.07
60-20-sf-3	-	-	-	60-20-sf-3	-	-	-	60-20-sf-3	-	-	-
60-20-sf-4	-	-	-	60-20-sf-4	3831.0	0.00	8.00	60-20-sf-4	3814.0	0.00	2.48
60-20-sf-5	-	-	-	60-20-sf-5	3651.0	0.00	2.71	60-20-sf-5	3637.0	0.00	2.59
60-30-sf-1	-	-	-	60-30-sf-1	6684.0	0.00	2.65	60-30-sf-1	6660.0	0.00	7.99
60-30-sf-2	-	-	-	60-30-sf-2	-	-	-	60-30-sf-2	5742.0	0.00	99.62
60-30-sf-3	-	-	-	60-30-sf-3	-	-	-	60-30-sf-3	5307.0	0.00	8.55
60-30-sf-4	-	-	-	60-30-sf-4	5526.0	0.00	135.77	60-30-sf-4	5496.0	0.00	19.20
60-30-sf-5	-	-	-	60-30-sf-5	-	-	-	60-30-sf-5	5293.0	0.00	7.50
90-10-sf-1	-	-	-	90-10-sf-1	-	-	-	90-10-sf-1	2921.0	0.00	4.38
90-10-sf-2	-	-	-	90-10-sf-2	-	-	-	90-10-sf-2	3002.0	0.00	0.83
90-10-sf-3	-	-	-	90-10-sf-3	2921.0	0.00	1.54	90-10-sf-3	2916.0	0.00	0.82
90-10-sf-4	-	-	-	90-10-sf-4	-	-	-	90-10-sf-4	-	-	-
90-10-sf-5	2639.0	0.00	8.43	90-10-sf-5	2626.0	0.00	8.07	90-10-sf-5	2617.0	0.00	4.76
90-20-sf-1	-	-	-	90-20-sf-1	6525.0	0.00	8.37	90-20-sf-1	6501.0	0.00	8.18
90-20-sf-2	-	-	-	90-20-sf-2	6808.0	0.00	9.36	90-20-sf-2	6786.0	0.00	8.01
90-20-sf-3	-	-	-	90-20-sf-3	6136.0	0.00	8.87	90-20-sf-3	6115.0	0.00	8.84
90-20-sf-4	-	-	-	90-20-sf-4	-	-	-	90-20-sf-4	5274.0	0.00	24.88
90-20-sf-5	-	-	-	90-20-sf-5	-	-	-	90-20-sf-5	5976.0	0.00	12.99
90-30-sf-1	-	-	-	90-30-sf-1	-	-	-	90-30-sf-1	-	-	-
90-30-sf-2	-	-	-	90-30-sf-2	-	-	-	90-30-sf-2	-	-	-
90-30-sf-3	-	-	-	90-30-sf-3	8671.0	1.15	182.44	90-30-sf-3	8647.0	0.00	94.60
90-30-sf-4	-	-	-	90-30-sf-4	9003.0	0.00	11.77	90-30-sf-4	8954.0	0.00	86.86
90-30-sf-5	-	-	-	90-30-sf-5	7993.0	0.00	10.52	90-30-sf-5	7973.0	0.00	35.25
120-10-sf-1	-	-	-	120-10-sf-1	-	-	-	120-10-sf-1	3855.0	0.00	9.00
120-10-sf-2	-	-	-	120-10-sf-2	-	-	-	120-10-sf-2	3555.0	0.00	8.66
120-10-sf-3	-	-	-	120-10-sf-3	-	-	-	120-10-sf-3	3413.0	0.00	2.32
120-10-sf-4	-	-	-	120-10-sf-4	4317.0	0.00	7.15	120-10-sf-4	4315.0	0.00	6.58
120-10-sf-5	-	-	-	120-10-sf-5	-	-	-	120-10-sf-5	3832.0	0.00	2.00
120-20-sf-1	8096.0	0.00	25.87	120-20-sf-1	8062.0	0.00	9.24	120-20-sf-1	8055.0	0.00	8.08
120-20-sf-2	-	-	-	120-20-sf-2	9326.0	0.00	11.35	120-20-sf-2	9282.0	0.00	7.10
120-20-sf-3	-	-	-	120-20-sf-3	-	-	-	120-20-sf-3	9633.0	0.00	37.84
120-20-sf-4	-	-	-	120-20-sf-4	-	-	-	120-20-sf-4	-	-	-
120-20-sf-5	-	-	-	120-20-sf-5	-	-	-	120-20-sf-5	-	-	-
120-30-sf-1	-	-	-	120-30-sf-1	13016.0	2.11	182.07	120-30-sf-1	12965.0	0.00	2.29
120-30-sf-2	-	-	-	120-30-sf-2	11937.0	0.84	183.83	120-30-sf-2	11900.0	0.00	103.62
120-30-sf-3	-	-	-	120-30-sf-3	11176.0	0.00	37.51	120-30-sf-3	11156.0	0.00	6.75
120-30-sf-4	-	-	-	120-30-sf-4	11438.0	0.00	186.12	120-30-sf-4	11396.0	0.00	181.72
120-30-sf-5	-	-	-	120-30-sf-5	-	-	-	120-30-sf-5	-	-	-

Table 12 Detailed Results of Ha26 on Standard-Weight Instances in S_A with sf execution mode ($ws = 0.7 - 0.9$)

ws-0.7				ws-0.8				ws-0.9			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-sf-1	1859.0	0.00	0.95	60-10-sf-1	1859.0	0.00	0.98	60-10-sf-1	1859.0	0.00	1.50
60-10-sf-2	1992.0	0.00	10.67	60-10-sf-2	1992.0	0.00	10.64	60-10-sf-2	1989.0	0.00	10.24
60-10-sf-3	2261.0	0.00	1.57	60-10-sf-3	2251.0	0.00	10.13	60-10-sf-3	2251.0	0.00	11.09
60-10-sf-4	2011.0	0.00	1.74	60-10-sf-4	2009.0	0.00	2.44	60-10-sf-4	2007.0	0.00	0.84
60-10-sf-5	1906.0	0.00	11.15	60-10-sf-5	1904.0	0.00	3.35	60-10-sf-5	1900.0	0.00	0.89
60-20-sf-1	3742.0	0.00	7.77	60-20-sf-1	3738.0	0.00	1.38	60-20-sf-1	3738.0	0.00	1.39
60-20-sf-2	3920.0	0.00	0.86	60-20-sf-2	3904.0	0.00	9.63	60-20-sf-2	3902.0	0.00	9.00
60-20-sf-3	3182.0	0.00	9.05	60-20-sf-3	3180.0	0.00	8.76	60-20-sf-3	3175.0	0.00	0.76
60-20-sf-4	3809.0	0.00	2.02	60-20-sf-4	3806.0	0.00	1.35	60-20-sf-4	3804.0	0.00	8.29
60-20-sf-5	3630.0	0.00	7.25	60-20-sf-5	3628.0	0.00	2.50	60-20-sf-5	3626.0	0.00	1.29
60-30-sf-1	6651.0	0.00	3.17	60-30-sf-1	6644.0	0.00	7.03	60-30-sf-1	6642.0	0.00	2.33
60-30-sf-2	5711.0	0.00	43.88	60-30-sf-2	5705.0	0.00	7.08	60-30-sf-2	5704.0	0.00	0.86
60-30-sf-3	5289.0	0.00	2.90	60-30-sf-3	5280.0	0.00	2.99	60-30-sf-3	5277.0	0.00	0.70
60-30-sf-4	5482.0	0.00	13.40	60-30-sf-4	5476.0	0.00	7.64	60-30-sf-4	5472.0	0.00	8.10
60-30-sf-5	5280.0	0.00	9.94	60-30-sf-5	5277.0	0.00	8.06	60-30-sf-5	5273.0	0.00	1.84
90-10-sf-1	2920.0	0.00	2.55	90-10-sf-1	2912.0	0.00	8.33	90-10-sf-1	2912.0	0.00	3.44
90-10-sf-2	2995.0	0.00	7.85	90-10-sf-2	2989.0	0.00	1.95	90-10-sf-2	2988.0	0.00	8.21
90-10-sf-3	2915.0	0.00	0.83	90-10-sf-3	2915.0	0.00	1.70	90-10-sf-3	2915.0	0.00	0.84
90-10-sf-4	-	-	-	90-10-sf-4	3153.0	0.00	0.75	90-10-sf-4	3153.0	0.00	9.16
90-10-sf-5	2613.0	0.00	2.25	90-10-sf-5	2612.0	0.00	0.80	90-10-sf-5	2612.0	0.00	9.17
90-20-sf-1	6491.0	0.00	4.78	90-20-sf-1	6487.0	0.00	0.73	90-20-sf-1	6487.0	0.00	7.42
90-20-sf-2	6784.0	0.00	7.03	90-20-sf-2	6781.0	0.00	7.16	90-20-sf-2	6775.0	0.00	1.27
90-20-sf-3	6109.0	0.00	9.20	90-20-sf-3	6107.0	0.00	5.77	90-20-sf-3	6107.0	0.00	5.34
90-20-sf-4	5262.0	0.00	12.59	90-20-sf-4	5255.0	0.00	7.71	90-20-sf-4	5252.0	0.00	7.53
90-20-sf-5	5966.0	0.00	8.52	90-20-sf-5	5955.0	0.00	7.71	90-20-sf-5	5952.0	0.00	2.01
90-30-sf-1	9535.0	0.00	49.69	90-30-sf-1	9502.0	0.00	2.50	90-30-sf-1	9498.0	0.00	1.81
90-30-sf-2	8506.0	0.00	10.70	90-30-sf-2	8495.0	0.00	7.18	90-30-sf-2	8485.0	0.00	1.32
90-30-sf-3	8638.0	0.00	33.32	90-30-sf-3	8634.0	0.00	5.81	90-30-sf-3	8628.0	0.00	1.56
90-30-sf-4	8935.0	0.00	9.20	90-30-sf-4	8931.0	0.00	1.39	90-30-sf-4	8929.0	0.00	0.71
90-30-sf-5	7963.0	0.00	1.55	90-30-sf-5	7961.0	0.00	2.53	90-30-sf-5	7959.0	0.00	0.72
120-10-sf-1	3852.0	0.00	8.03	120-10-sf-1	3850.0	0.00	0.70	120-10-sf-1	3850.0	0.00	9.11
120-10-sf-2	3554.0	0.00	1.54	120-10-sf-2	3551.0	0.00	1.39	120-10-sf-2	3551.0	0.00	1.37
120-10-sf-3	3410.0	0.00	4.53	120-10-sf-3	3409.0	0.00	8.53	120-10-sf-3	3409.0	0.00	2.77
120-10-sf-4	4311.0	0.00	7.84	120-10-sf-4	4307.0	0.00	4.94	120-10-sf-4	4289.0	0.00	0.66
120-10-sf-5	3828.0	0.00	1.87	120-10-sf-5	3821.0	0.00	1.32	120-10-sf-5	3821.0	0.00	8.19
120-20-sf-1	8051.0	0.00	1.43	120-20-sf-1	8043.0	0.00	1.28	120-20-sf-1	8043.0	0.00	7.14
120-20-sf-2	9264.0	0.00	6.67	120-20-sf-2	9255.0	0.00	6.80	120-20-sf-2	9246.0	0.00	4.17
120-20-sf-3	9610.0	0.00	39.06	120-20-sf-3	9596.0	0.00	3.40	120-20-sf-3	9590.0	0.00	1.38
120-20-sf-4	-	-	-	120-20-sf-4	7697.0	0.00	7.99	120-20-sf-4	7695.0	0.00	6.85
120-20-sf-5	-	-	-	120-20-sf-5	7455.0	0.00	1.27	120-20-sf-5	7455.0	0.00	0.71
120-30-sf-1	12956.0	0.00	0.96	120-30-sf-1	12954.0	0.00	0.89	120-30-sf-1	12954.0	0.00	1.26
120-30-sf-2	11889.0	0.00	181.30	120-30-sf-2	11884.0	0.00	4.43	120-30-sf-2	11879.0	0.00	13.40
120-30-sf-3	11152.0	0.00	2.55	120-30-sf-3	11152.0	0.00	0.80	120-30-sf-3	11152.0	0.00	1.36
120-30-sf-4	11368.0	0.00	3.92	120-30-sf-4	11366.0	0.00	2.83	120-30-sf-4	11357.0	0.00	1.32
120-30-sf-5	13293.0	0.00	16.81	120-30-sf-5	13287.0	0.00	3.46	120-30-sf-5	13266.0	0.00	1.92

2.3. Results on the instances with sif execution mode

Table 13 Detailed Results of Ha26 on Standard-Weight Instances in S_A with sif execution mode ($ws = 0.4 - 0.6$)

ws-0.4				ws-0.5				ws-0.6			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-sif-1	1880.0	0.00	1.28	60-10-sif-1	1872.0	0.00	6.30	60-10-sif-1	1866.0	0.00	9.67
60-10-sif-2	-	-	-	60-10-sif-2	2021.0	0.00	1.59	60-10-sif-2	2005.0	0.00	10.21
60-10-sif-3	2289.0	0.00	3.79	60-10-sif-3	2271.0	0.00	9.43	60-10-sif-3	2268.0	0.00	7.82
60-10-sif-4	-	-	-	60-10-sif-4	2037.0	0.00	0.89	60-10-sif-4	2019.0	0.00	7.91
60-10-sif-5	1917.0	0.00	0.92	60-10-sif-5	1912.0	0.00	8.06	60-10-sif-5	1908.0	0.00	6.96
60-20-sif-1	-	-	-	60-20-sif-1	3757.0	0.00	4.92	60-20-sif-1	3748.0	0.00	10.41
60-20-sif-2	-	-	-	60-20-sif-2	3939.0	0.00	15.09	60-20-sif-2	3926.0	0.00	12.03
60-20-sif-3	-	-	-	60-20-sif-3	-	-	-	60-20-sif-3	3196.0	0.00	8.45
60-20-sif-4	-	-	-	60-20-sif-4	3826.0	0.00	3.31	60-20-sif-4	3814.0	0.00	9.78
60-20-sif-5	-	-	-	60-20-sif-5	3652.0	0.00	8.06	60-20-sif-5	3639.0	0.00	7.92
60-30-sif-1	-	-	-	60-30-sif-1	6674.0	0.00	11.27	60-30-sif-1	6662.0	0.00	1.46
60-30-sif-2	-	-	-	60-30-sif-2	-	-	-	60-30-sif-2	5744.0	0.00	147.67
60-30-sif-3	-	-	-	60-30-sif-3	-	-	-	60-30-sif-3	5298.0	0.00	10.42
60-30-sif-4	-	-	-	60-30-sif-4	-	-	-	60-30-sif-4	5492.0	0.00	8.92
60-30-sif-5	-	-	-	60-30-sif-5	-	-	-	60-30-sif-5	5297.0	0.00	12.87
90-10-sif-1	-	-	-	90-10-sif-1	2934.0	0.00	18.02	90-10-sif-1	2925.0	0.00	9.16
90-10-sif-2	-	-	-	90-10-sif-2	3017.0	0.00	3.68	90-10-sif-2	2998.0	0.00	8.61
90-10-sif-3	-	-	-	90-10-sif-3	2930.0	0.00	9.53	90-10-sif-3	2917.0	0.00	0.84
90-10-sif-4	-	-	-	90-10-sif-4	-	-	-	90-10-sif-4	-	-	-
90-10-sif-5	-	-	-	90-10-sif-5	2626.0	0.00	8.91	90-10-sif-5	2617.0	0.00	4.76
90-20-sif-1	-	-	-	90-20-sif-1	6538.0	0.00	4.66	90-20-sif-1	6501.0	0.00	11.24
90-20-sif-2	-	-	-	90-20-sif-2	6839.0	0.00	80.55	90-20-sif-2	6800.0	0.00	35.57
90-20-sif-3	-	-	-	90-20-sif-3	6140.0	0.00	23.71	90-20-sif-3	6117.0	0.00	14.80
90-20-sif-4	-	-	-	90-20-sif-4	-	-	-	90-20-sif-4	5273.0	0.00	15.32
90-20-sif-5	-	-	-	90-20-sif-5	-	-	-	90-20-sif-5	5976.0	0.00	8.44
90-30-sif-1	-	-	-	90-30-sif-1	9596.0	0.00	186.68	90-30-sif-1	9551.0	1.92	181.54
90-30-sif-2	-	-	-	90-30-sif-2	-	-	-	90-30-sif-2	8524.0	0.00	25.30
90-30-sif-3	-	-	-	90-30-sif-3	8673.0	0.00	21.55	90-30-sif-3	8648.0	0.00	17.94
90-30-sif-4	-	-	-	90-30-sif-4	8986.0	0.00	150.12	90-30-sif-4	8947.0	0.00	15.17
90-30-sif-5	8049.0	0.00	3.68	90-30-sif-5	7992.0	0.00	52.16	90-30-sif-5	7971.0	0.00	62.93
120-10-sif-1	-	-	-	120-10-sif-1	-	-	-	120-10-sif-1	3868.0	0.00	7.85
120-10-sif-2	3567.0	0.00	8.11	120-10-sif-2	3557.0	0.00	3.18	120-10-sif-2	3555.0	0.00	0.71
120-10-sif-3	-	-	-	120-10-sif-3	-	-	-	120-10-sif-3	3416.0	0.00	8.13
120-10-sif-4	-	-	-	120-10-sif-4	4319.0	0.00	7.60	120-10-sif-4	4307.0	0.00	6.77
120-10-sif-5	-	-	-	120-10-sif-5	-	-	-	120-10-sif-5	3834.0	0.00	5.50
120-20-sif-1	-	-	-	120-20-sif-1	8066.0	0.00	9.60	120-20-sif-1	8055.0	0.00	10.29
120-20-sif-2	-	-	-	120-20-sif-2	9318.0	0.00	10.09	120-20-sif-2	9276.0	0.00	5.68
120-20-sif-3	-	-	-	120-20-sif-3	-	-	-	120-20-sif-3	9643.0	0.00	50.28
120-20-sif-4	-	-	-	120-20-sif-4	7741.0	0.00	22.45	120-20-sif-4	7725.0	0.00	13.40
120-20-sif-5	-	-	-	120-20-sif-5	-	-	-	120-20-sif-5	7472.0	0.00	18.01
120-30-sif-1	-	-	-	120-30-sif-1	13058.0	4.79	181.26	120-30-sif-1	12975.0	0.00	181.82
120-30-sif-2	-	-	-	120-30-sif-2	11924.0	0.00	182.26	120-30-sif-2	11898.0	0.00	96.80
120-30-sif-3	-	-	-	120-30-sif-3	11195.0	0.00	68.13	120-30-sif-3	11157.0	0.00	2.93
120-30-sif-4	-	-	-	120-30-sif-4	11454.0	4.43	183.46	120-30-sif-4	11390.0	0.00	106.56
120-30-sif-5	-	-	-	120-30-sif-5	-	-	-	120-30-sif-5	13316.0	0.00	183.86

Table 14 Detailed Results of Ha26 on Standard-Weight Instances in S_A with sif execution mode ($ws = 0.7 - 0.9$)

ws-0.7				ws-0.8				ws-0.9			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
60-10-sif-1	1862.0	0.00	9.24	60-10-sif-1	1859.0	0.00	0.97	60-10-sif-1	1859.0	0.00	1.50
60-10-sif-2	2003.0	0.00	8.69	60-10-sif-2	1992.0	0.00	4.52	60-10-sif-2	1989.0	0.00	10.36
60-10-sif-3	2263.0	0.00	9.01	60-10-sif-3	2260.0	0.00	10.98	60-10-sif-3	2251.0	0.00	10.87
60-10-sif-4	2011.0	0.00	5.92	60-10-sif-4	2009.0	0.00	2.46	60-10-sif-4	2007.0	0.00	0.85
60-10-sif-5	1904.0	0.00	1.66	60-10-sif-5	1902.0	0.00	0.72	60-10-sif-5	1900.0	0.00	0.87
60-20-sif-1	3742.0	0.00	5.72	60-20-sif-1	3738.0	0.00	0.78	60-20-sif-1	3738.0	0.00	2.00
60-20-sif-2	3919.0	0.00	8.70	60-20-sif-2	3912.0	0.00	9.14	60-20-sif-2	3902.0	0.00	8.67
60-20-sif-3	3178.0	0.00	0.93	60-20-sif-3	3178.0	0.00	1.70	60-20-sif-3	3175.0	0.00	0.79
60-20-sif-4	3811.0	0.00	7.92	60-20-sif-4	3806.0	0.00	1.40	60-20-sif-4	3804.0	0.00	8.41
60-20-sif-5	3630.0	0.00	6.70	60-20-sif-5	3628.0	0.00	2.47	60-20-sif-5	3626.0	0.00	1.25
60-30-sif-1	6651.0	0.00	9.36	60-30-sif-1	6639.0	0.00	5.52	60-30-sif-1	6636.0	0.00	1.24
60-30-sif-2	5714.0	0.00	180.89	60-30-sif-2	5705.0	0.00	3.78	60-30-sif-2	5704.0	0.00	0.69
60-30-sif-3	5289.0	0.00	13.69	60-30-sif-3	5283.0	0.00	1.47	60-30-sif-3	5277.0	0.00	1.27
60-30-sif-4	5481.0	0.00	14.60	60-30-sif-4	5476.0	0.00	10.05	60-30-sif-4	5472.0	0.00	0.70
60-30-sif-5	5283.0	0.00	35.05	60-30-sif-5	5277.0	0.00	8.48	60-30-sif-5	5273.0	0.00	1.30
90-10-sif-1	2917.0	0.00	5.59	90-10-sif-1	2915.0	0.00	4.46	90-10-sif-1	2907.0	0.00	9.00
90-10-sif-2	2992.0	0.00	8.03	90-10-sif-2	2989.0	0.00	7.82	90-10-sif-2	2988.0	0.00	8.26
90-10-sif-3	2915.0	0.00	0.80	90-10-sif-3	2915.0	0.00	0.95	90-10-sif-3	2915.0	0.00	0.85
90-10-sif-4	3160.0	0.00	2.71	90-10-sif-4	3159.0	0.00	8.58	90-10-sif-4	3153.0	0.00	9.35
90-10-sif-5	2616.0	0.00	5.81	90-10-sif-5	2612.0	0.00	0.77	90-10-sif-5	2612.0	0.00	9.25
90-20-sif-1	6491.0	0.00	7.48	90-20-sif-1	6487.0	0.00	1.35	90-20-sif-1	6487.0	0.00	2.07
90-20-sif-2	6784.0	0.00	8.09	90-20-sif-2	6781.0	0.00	7.88	90-20-sif-2	6775.0	0.00	1.27
90-20-sif-3	6110.0	0.00	7.37	90-20-sif-3	6101.0	0.00	1.87	90-20-sif-3	6101.0	0.00	2.44
90-20-sif-4	5261.0	0.00	8.67	90-20-sif-4	5255.0	0.00	14.46	90-20-sif-4	5252.0	0.00	4.22
90-20-sif-5	5962.0	0.00	8.15	90-20-sif-5	5955.0	0.00	7.96	90-20-sif-5	5952.0	0.00	2.06
90-30-sif-1	9529.0	0.00	21.26	90-30-sif-1	9523.0	0.00	5.28	90-30-sif-1	9498.0	0.00	7.59
90-30-sif-2	8502.0	0.00	30.65	90-30-sif-2	8494.0	0.00	8.41	90-30-sif-2	8491.0	0.00	3.65
90-30-sif-3	8640.0	0.00	47.97	90-30-sif-3	8634.0	0.00	9.01	90-30-sif-3	8628.0	0.00	1.73
90-30-sif-4	8935.0	0.00	2.26	90-30-sif-4	8931.0	0.00	1.54	90-30-sif-4	8929.0	0.00	1.38
90-30-sif-5	7963.0	0.00	2.28	90-30-sif-5	7961.0	0.00	3.68	90-30-sif-5	7959.0	0.00	1.02
120-10-sif-1	3866.0	0.00	7.89	120-10-sif-1	3850.0	0.00	0.73	120-10-sif-1	3850.0	0.00	1.38
120-10-sif-2	3552.0	0.00	1.42	120-10-sif-2	3549.0	0.00	0.74	120-10-sif-2	3549.0	0.00	0.74
120-10-sif-3	3408.0	0.00	8.56	120-10-sif-3	3407.0	0.00	0.80	120-10-sif-3	3407.0	0.00	0.75
120-10-sif-4	4307.0	0.00	7.49	120-10-sif-4	4305.0	0.00	6.54	120-10-sif-4	4289.0	0.00	0.68
120-10-sif-5	3823.0	0.00	2.14	120-10-sif-5	3819.0	0.00	1.87	120-10-sif-5	3819.0	0.00	7.54
120-20-sif-1	8051.0	0.00	1.45	120-20-sif-1	8049.0	0.00	1.36	120-20-sif-1	8043.0	0.00	1.26
120-20-sif-2	9264.0	0.00	7.17	120-20-sif-2	9255.0	0.00	0.70	120-20-sif-2	9246.0	0.00	4.15
120-20-sif-3	9618.0	0.00	14.71	120-20-sif-3	9598.0	0.00	2.58	120-20-sif-3	9590.0	0.00	1.22
120-20-sif-4	7718.0	0.00	9.00	120-20-sif-4	7716.0	0.00	3.20	120-20-sif-4	7695.0	0.00	0.76
120-20-sif-5	7460.0	0.00	14.14	120-20-sif-5	7455.0	0.00	2.59	120-20-sif-5	7455.0	0.00	0.73
120-30-sif-1	12959.0	0.00	15.32	120-30-sif-1	12954.0	0.00	0.99	120-30-sif-1	12954.0	0.00	0.92
120-30-sif-2	11887.0	0.00	180.83	120-30-sif-2	11884.0	0.00	6.62	120-30-sif-2	11879.0	0.00	12.47
120-30-sif-3	11154.0	0.00	9.04	120-30-sif-3	11152.0	0.00	6.91	120-30-sif-3	11152.0	0.00	0.88
120-30-sif-4	11368.0	0.00	180.95	120-30-sif-4	11366.0	0.00	7.63	120-30-sif-4	11366.0	0.00	3.28
120-30-sif-5	13287.0	0.00	49.29	120-30-sif-5	13281.0	0.00	8.43	120-30-sif-5	13281.0	0.00	3.81